

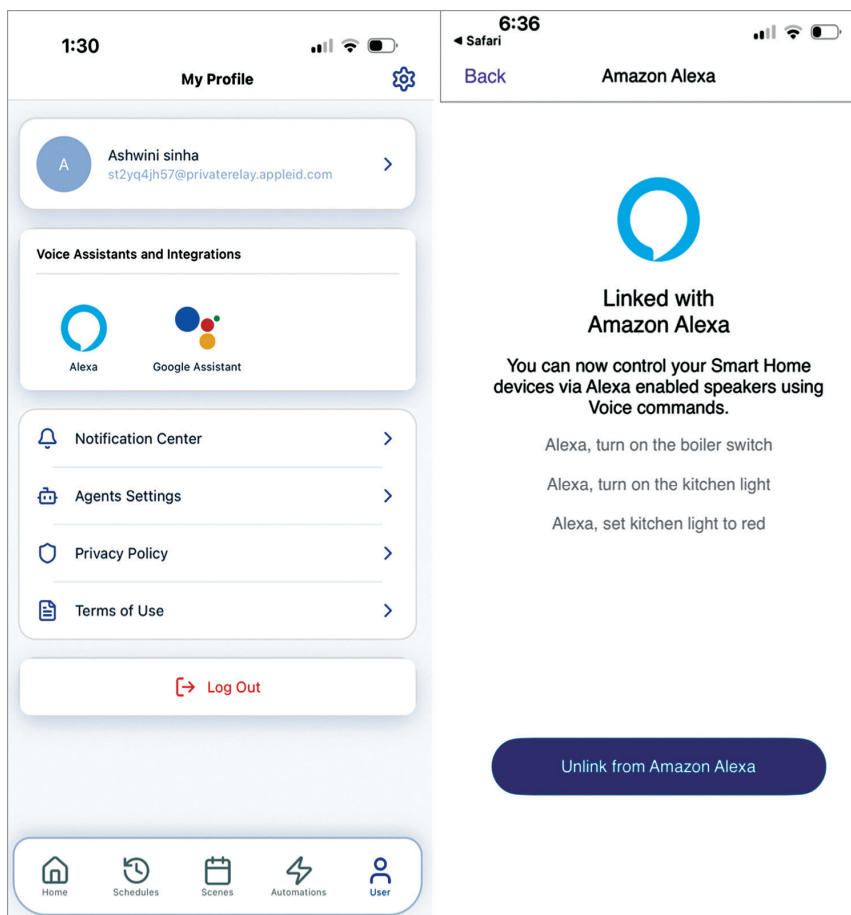


Fig. 10: Device integrated with Alexa and Google Assistant for voice control and automation

The button can be triggered through the app interface or via Alexa or other AI assistants by requesting a device sync. Otherwise, the application automatically synchronises data at one-hour intervals, which can be adjusted as required. After completing configuration and calibration, upload the code to the board. Fig. 7 shows the RainMaker code snippet with calibration parameters and sensor pin configuration.

Provisioning the device

After uploading the firmware, open the serial monitor. A link for generating the provisioning QR code is displayed. Copy and paste this link into a web browser to generate the QR code, then save and download it.

For production deployment, the QR code should be printed on the manual or packaging and supplied to the customer. The device can

then be provisioned by scanning the QR code. Fig. 8 shows the generated provisioning code.

Configuring the app

Download the RainMaker app from the App Store or Play Store and sign in using an email or iCloud account. Tap the plus (+) icon and scan the saved QR code. Once connected to the device's Wi-Fi network, the app prompts selection of the home network and entry of the password.

Adding device to AI assistant

The device supports integration with Alexa, Google Assistant, and Apple Home. After linking, appliances can be controlled through voice commands or automated using sensor data and other connected devices.

To enable integration, open User Options in the RainMaker app

TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to <https://sourcecode.electronicsforu.com> to access and download source code and other resources for this project

and select the preferred assistant. Follow the on-screen instructions to complete the linking process. After successful linking, the device appears automatically in the assistant's device list, where voice commands and automation routines can be configured.

Testing

Testing the smart energy meter with Alexa is straightforward. Issuing the command "Alexa, turn on Sync on the Energy Meter" updates the device and sends the latest sensor data to the cloud. Queries such as "What is my electricity bill?" or "How much power is the appliance using?" return real-time responses.

Alternatively, the RainMaker Home app allows manual synchronisation using the Sync button. Detailed readings, including voltage, current, power consumption, total energy (kWh), and estimated billing, are available in the Energy Monitor section.

The two relay switches function as standard on/off controls, enabling remote operation of connected appliances. The meter also provides built-in protection and seamless voice control through Alexa, Google Home, or Apple HomeKit. The left side of Fig. 1 illustrates overall device control and monitoring, while its right side displays real-time voltage, current, power, energy consumption (kWh), and estimated billing data. **EFY**

Ashwini Kumar Sinha, an IoT and AI enthusiast, is Tech Journalist at EFY.